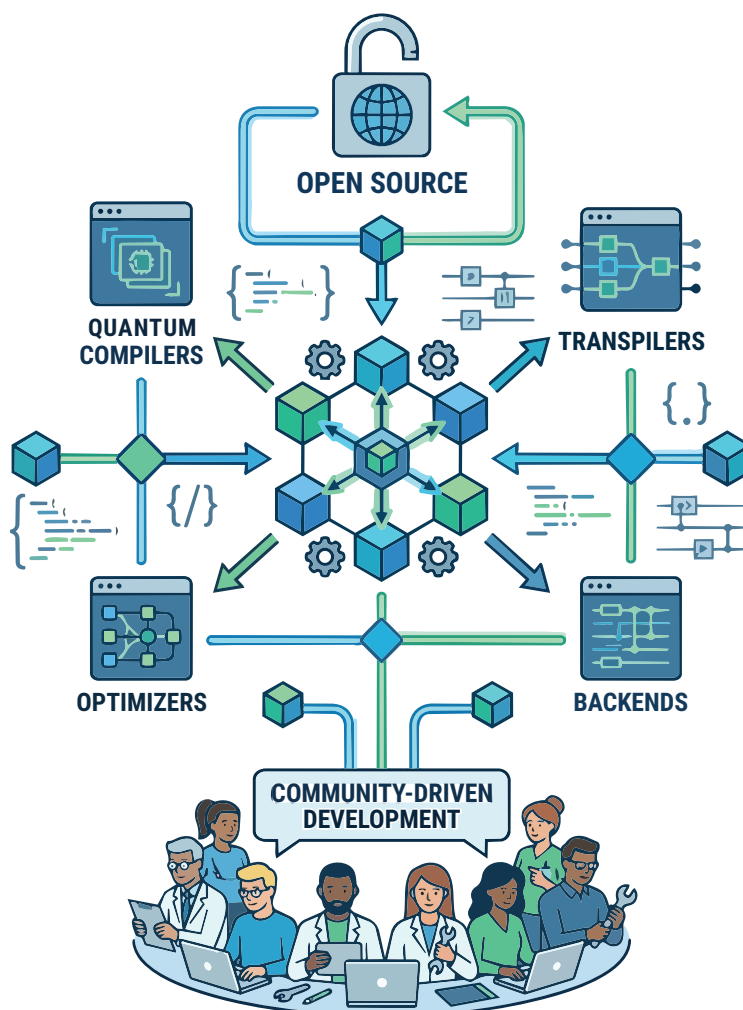


Open Source Quantum Compilers: A Technical Overview of the Current Landscape

Quality quantum compilers are a necessity to get the best results from noisy quantum machines. Fortunately, there are quite a few open source compilers that are up to the task.



Every quantum program needs to transform abstract logic of qubits and gates into a sequence of control signals a machine can interpret. Software that carries out this function is called a quantum compiler. Constructing a compiler for early-stage quantum machines that have limited connectivity, coherence, and suffer from noise is a major challenge for quantum computing. For such machines, the quality of a compiler can

make a considerable difference in the performance of an algorithm.

In classical computing, the main challenges are speed and code size, whereas in quantum computing, the main problems are decoherence, gate errors, and qubit connectivity. Despite these problems, hardware vendors have been sustaining an active open source community of quantum compiler builders. Vendor competition has made the quality of available quantum compilers a priority.

What a quantum compiler does

A quantum compiler lowers a circuit from a high level of abstraction to a level that a machine can execute. The stages in this process include:

- Circuit translation — Conversion of a circuit to an internal intermediate representation, which is typically a directed acyclic graph (DAG) of gates.
- Circuit optimisation — Combining, cancelling, and rewriting gate